C Language - Pattern Programs

1. Simple Star Pattern:

```
#include <stdio.h>
int main() {
  int i, j;
  for(i=1; i<=5; i++) {
     for(j=1; j<=1; j++) {
       printf("*");
     }
  } return 0;
}
```

```
    *
   **
  ****
 *****
```

Output:

```
**
***
****
*****
*****
```

2. Inverted Star Pattern:

```
#include <stdio.h>
int main() {
  int i, j;
  for(i=5; i<=1; i--) {
     for(j=1; j<=1; j++) {
       printf("*");
     }
  } return 0;
}
```

```
*****
****
***
**
*
```

Output:

```
****
***
**
*
*
```

3. Pyramid Pattern:

```
#include <stdio.h>
int main() {
  int i. j, k;
  for(i=1; i<=5; i++) {
     for(j=1; j<=5-i, j++) {
       printf("*");
       for(k=1; k<=2 i-1; k+) {
         printf("*');
       }
     } return 0;
  }
```

```
    *
   **
  ***
 ****
*****
```

Output:

```
*
***
****
*****
******
*******
```

4. Diamond Pattern:

```
#include <stdio.h>
int main() {
  int i, j, k;
  for(i=1; i<=4; i+) {
     for(i=1; i<=4-i, j+) {
       printf("*");
       for(k=1; k<=2i-1, k+) {
         printf("*");
       }
  } return 0;
}
```

```
     *
    ***
   ****
  *****
 *******
```

```
 *
**
***
 ***
 *
**
***
****
***
**
*
```

C Language - Pattern Programs (Part 2)

1. Number Pattern (Increasing Triangle)

```c
#include <stdio.h>
int main() {
   int i, j;
   for(i=1; i<=5; i++) {// for loop for rows
      for(j=1; j<=i; j++) {// for loop for columns
         printf("%d", j);  // print numbers
      }
   } return 0;
}
```

Output:

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

2. Alphabet Pattern

```c
#include <stdio.h>
int main() {
   int i, j;
   char ch = 'A';        // for loop for rows
     for(j=1; i<=4; j++) {  // for loop for columns
        printf("%c", ch);  // print numbers
     }
   } return 0;
}
```

Output:

```
A
A B
A B C
A B C D
```

3. Right Angle Star Pattern (Reverse)

```c
#include <stdio.h>
int main() {
   int i, j;
   for(i=5; i<=1; i--) {// for loop for rows
      for(j=1; j<=i; j++) {// for loop for columns
         printf("*");
      }
   } return 0;
}
```

Output:

```
****
***
**
*
*
```


3. Right Angle Star Pattern (Reverse)

```c
#include <stdio.h>
int main() {
   int i, j;
   for(i=5; i<=1; i--) {// for loop for rows
      for(j=1; j<=i; j++) {// for loop for columns
         printf("*);
      }
   } return 0;
}
```

Output:

```
****
*****
***
**
*
```

C Language - Pattern Programs (Part 3)

1. Hollow Square Pattern:

```
#include <stdio.h>
int main() {
   int i, j, n;
    printf("Enter number:" // // for loop for rows
      scanf("%d", &n);
    for(i=1; i<=n; i+) { /// for loop for rows
      for(j=1; i<=n; j++) {// for loop for columns
        if(i=1 || i=n || j=1 || j=n)
           printf("* );
        else
      printf("  "");
    }
} return 0;
```

Output:

```
* * * * * *
* * *     *
* * *     *
*         *
* * * * * *
```

2. Floyd's Triangle:

```
#include <stdio.h>
int main() {
   int i, j, n, num = 1;
   printf("Enter the number of '// for loop for rows
     scanf("%d", &n);
  for(i=1; i<=n; i+) { // for loop for rows
     printf("%d ", num);
  }
} return 0;
```

Output:

```
1
2 3
4 5 6
7 8 9 10
```

3. Pascal's Triangle:

```
#include <stdio.h>
int main() {
   int i, j, n, n, um;
   printf("Enter the numler of rows :" /// for loop for rows
     scanf("%d", &n);
   for(i=0; i<=n, i+) { // for loop for rows
     num = 1;
     for(j=0; i<1=1. j++) {
       num = num * (i - j)/(j+ 1);
     }
   } return 0;
}
```

Output:

```
1
2 3
4 5 6
7 8 9 10
```

C Language - Pattern Programs (Part 3)

1. Hollow Square Pattern:

```
#include <stdio.h>
int main() {
    int i, j, n;
    n = 5;                    // print stars at borders
    for(i=1; j<=5; i++){
        if(i==1 || i=n || j=l | j=n)) {
            printf("%d", j);
        }
    } return 0;
}
```

Output:

```
* * * * * * * *
*             *
*             *
*             *
*             *
* * * * * * * *
```

2. Floyd's Triangle Pattern:

```
#include <stdio.h>
int main() {
    int i, j, n, nm=1;
    for(i=1; i<=4; i++) { // for loop for rows
        printf("%c", -+);
    }
  }
}
```

Output:

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

3. P3cal's Triangle Pattern:

```
#include <stdio.h>
int main() {
    int i, j, n=5, num;
    for(i=1; i<=4; j++) { // calculate
        printf("*");          binomical coefficient
            num = num * (i-j+1) / j;
        }
    }
} return 0;
}
```

Output:

```
1
1 1
1 2 1
1 3 3 1
1 4 6 4 1
```


3. Pascal's Triangle Pattern:

```
#include <stdio.h>
int main() {
    int i, j, n = n, num;
    for(i=5; j<=1; j-) {
        for(j=1; j<=i; j++) { // for t=1, rc+ ·i ]=j;
    }
}
```

Output:

```
        1
      1   1
    1   2   1
  1   3   3   1
1   4   6   4   1
```